Intrusion Detection System using Machine Learning with Extended Analysis on NSL-KDD Dataset

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Abstract

Intrusion Detection Systems (IDS) play a vital role in securing computer networks by detecting malicious activities. Traditional IDS techniques rely on rule-based approaches, which often fail to detect new and unknown attacks. This paper presents a machine learning-based approach for intrusion detection using the NSL-KDD dataset. Various supervised learning algorithms such as K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest (RF), Naive Bayes (NB), Logistic Regression (LR), and Decision Tree (DT) are implemented and compared. Furthermore, this work extends the original study by incorporating feature scaling, feature selection, cross-validation, and performance visualization techniques. The results demonstrate that the Decision Tree classifier achieves the highest accuracy among all models. The study highlights the effectiveness of machine learning techniques in improving intrusion detection systems.

➢ Introduction

With the rapid growth of internet usage, cyber attacks have become increasingly common. Protecting sensitive information from unauthorized access is a major concern in cybersecurity. Intrusion Detection Systems (IDS) are designed to monitor network traffic and identify suspicious activities.

Traditional IDS methods are primarily rule-based and depend on predefined signatures. These systems are ineffective in detecting new or unknown attacks and often produce high false alarm rates. Machine learning techniques provide a better solution by learning patterns from data and detecting anomalies more accurately.

This paper focuses on implementing and comparing multiple machine learning algorithms for intrusion detection using the NSL-KDD dataset.

➢ Related Work

Several studies have explored the use of machine learning in intrusion detection systems. Previous research applied algorithms such as KNN, SVM, and Naive Bayes on datasets like KDD Cup 1999 and NSL-KDD. These studies demonstrated that machine learning techniques can improve detection accuracy.

However, many existing works lack proper preprocessing, feature selection, and performance evaluation techniques. This study extends previous work by incorporating additional techniques to enhance model performance.

➢ Dataset Description

The NSL-KDD dataset is an improved version of the KDD Cup 1999 dataset and is widely used for evaluating intrusion detection systems. It eliminates redundant records and provides a balanced dataset.

The dataset consists of:

- 41 input features
- 1 output label (normal or attack)

Attack types include:

- Denial of Service (DoS)
- Probe
- Remote to Local (R2L)
- User to Root (U2R)

The dataset is divided into training and testing sets for model evaluation.

➢ Methodology

The methodology used in this study consists of the following steps:

❖ **Data Preprocessing**

- Loaded dataset into Python environment
- Handled categorical features using encoding
- Converted labels into binary classification (Normal vs Attack)

❖ **Feature Scaling**

- Applied MinMaxScaler to normalize feature values
- Ensures all values lie between 0 and 1

❖ **Feature Selection**

- Applied Chi-Square test to select important features
- Reduced dimensionality and improved model performance

❖ **Model Training**

The following supervised learning algorithms were used:

- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Random Forest (RF)
- Naive Bayes (NB)
- Logistic Regression (LR)
- Decision Tree (DT)

❖ **Model Evaluation**

- Accuracy score
- Confusion matrix
- ROC curve
- Cross-validation

➢ Results and Discussion

The performance of different models is shown below:

Model	Accuracy
KNN	0.7925
SVM	0.7815
Random Forest	0.7732
Naive Bayes	0.7547
Logistic Regression	0.7603
Decision Tree	**0.8007**

The Decision Tree model achieved the highest accuracy among all models. KNN also performed well but had higher computational time. Naive Bayes showed the lowest performance.

Graphs such as accuracy comparison, confusion matrix, and ROC curves were used for better visualization and analysis.

➢ Proposed Extension

This work extends the original study by incorporating the following improvements:

- Feature scaling using MinMaxScaler
- Feature selection using Chi-square method
- Cross-validation for better model reliability
- Comparison of multiple machine learning models
- Visualization using graphs

These extensions improved the overall performance and reliability of the intrusion detection system.

➢ Conclusion

This study demonstrates the effectiveness of machine learning algorithms in intrusion detection systems. Among all models, the Decision Tree classifier achieved the best performance. Feature selection and scaling significantly improved model accuracy.

The results indicate that machine learning-based IDS can effectively detect network intrusions and enhance cybersecurity systems.

References

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